

(58 & A-29)

SEAT No. _____

No. of Printed Pages : 2

SARDAR PATEL UNIVERSITY

B.C.A. – VI SEMESTER (CBCS)

US06EBCA01 : Information Security

Date : 01/04/2017, Saturday Time : 2:00 PM to 4:00 PM

Max Marks : 70

Q:1 Write answers of following Multiple Choice Questions :

[10]

- [01] Which action is that compromises the security of information owned by an organization?
(A) Security attack (B) Service attack
(C) Threat (D) None of these
- [02] Which one is the category of the active attack?
(A) Traffic analysis (B) Masquerade
(C) Passive attack (D) Release of message contents
- [03] Cryptography is also known as _____.
(A) Plain text writing (B) Stream writing
(C) Secret writing (D) None of these
- [04] The ____ is the original message or data that is fed into the algorithm as input.
(A) Secret key (B) Plaintext
(C) Encryption algorithm (D) Ciphertext
- [05] The process of attempting to discover the plaintext or key is known as _____.
(A) Stream cipher (B) Cryptanalysis
(C) Cipher text (D) Block cipher
- [06] A _____ is a program or a part of program that are activated by a trigger.
(A) Worms (B) Virus
(C) Trojan horses (D) Logic bomb
- [07] What the virus does besides spreading is known as _____.
(A) Payload (B) Triggering
(C) Dormant (D) Infection vector
- [08] The ____ is a person who authorized for access but misuses his/her privileges.
(A) Misfeasor (B) Masquerader
(C) Clandestine user (D) None of these
- [09] The _____ server is a server that creates a secure tunnel connection.
(A) Authentication (B) VPN
(C) Tunnel (D) DNS
- [10] The _____ controls how particular services are used.
(A) User control (B) Service control
(C) Behavior control (D) Direction control

Q:2 Answer the following short questions : Attempt Any Ten [20]

- [01] What is masquerade and replay?
- [02] What do you mean by security mechanism and security services?
- [03] What do you mean by access control and data confidentiality?
- [04] Define plain text and cipher text.
- [05] What are the principles of Symmetric/conventional encryption?
- [06] List types of attacks on encrypted message.
- [07] What do you mean by misfeasor?
- [08] Define : Logic bomb
- [09] Write triggering and execution phases during virus lifetime.
- [10] List types of firewall.
- [11] What do you mean by application level gateway?
- [12] What is circuit level gateway?

Q:3 [A] What is security attack? Explain passive attack in detail. [06]

[B] Explain categories of security services. [04]

OR

Q:3 [C] Explain active attack in detail. [06]

[D] Explain the network security model in brief. [04]

Q:4 [A] What is cryptography? Explain symmetric key and asymmetric key cryptography. [06]

[B] Explain public - key cryptography in detail. [04]

OR

Q:4 [C] Explain substitution and transposition cipher with example. [06]

[D] Explain the applications for public - key cryptosystems. [04]

Q:5 [A] Classify the viruses in detail. [05]

[B] Write a short note on Backdoor. [05]

OR

Q:5 [C] Explain different nature of viruses in detail. [05]

[D] Write a note on Intruders. [05]

Q:6 [A] What is Firewall? Explain packet filtering firewall in detail. [10]

OR

Q:6 [B] Explain digital signature process. Write comparison between conventional and digital signature. [10]